

CASE 10

ACQUIRED IMMUNE DEFICIENCY SYNDROME (AIDS)

Infection can suppress adaptive immunity.

Certain infectious microorganisms can suppress or subvert the immune system. For example, in lepromatous leprosy, *Mycobacterium leprae* induces T cells to produce lymphokines that stimulate a humoral response but suppress the development of a successful inflammatory response to contain the leprosy bacillus. The leprosy bacillus multiplies and there is a persistent depression of cell-mediated immune responses to a wide range of antigens (see Case 46). Another example of immunosuppression is provided by bacterial superantigens, such as toxic shock syndrome toxin-1. Superantigens bind and stimulate large numbers of T cells by binding to certain $V\beta$ chains of the T-cell receptor, inducing massive production of cytokines by the responding T cells (see Case 45). This, in turn, causes a temporary suppression of adaptive immunity.

At the beginning of the 20th century, when tuberculosis was the leading cause of death and fully half the population was tuberculin-positive, it was well known that an intercurrent measles infection would cause a well-contained tuberculosis infection to run rampant and result in death. The mechanism responsible is now known to be the suppression of IL-2 synthesis after binding of measles virus to CD46 on macrophages.

Some of the microorganisms that suppress immunity act by infecting lymphocytes. Infectious mononucleosis or glandular fever is caused by a virus (Epstein-Barr virus) that infects B lymphocytes. The infection activates cytotoxic CD8 T cells, which destroy the B cells in which the Epstein-Barr virus is replicating. In the third week of infection, at the height of activation of CD8 T cells, all adaptive immunity is suppressed. The cytokines responsible for the immunosuppression are not well defined but probably include IL-10 and TGF- β (see Case 43).

The human immunodeficiency virus (HIV) presents a chilling example of the consequences of infection and destruction of immune cells by a microorganism. CD4 molecules on the T-cell surface act as the receptors for HIV (Fig. 10.1). CD4 is also expressed on the surface of cells of the macrophage lineage, and they, too, can be

TOPICS BEARING ON THIS CASE:

Failure of cell-mediated immunity

Infection with the human immunodeficiency virus (HIV)

Control of HIV infection

Drug therapy for HIV infection

ELISA test

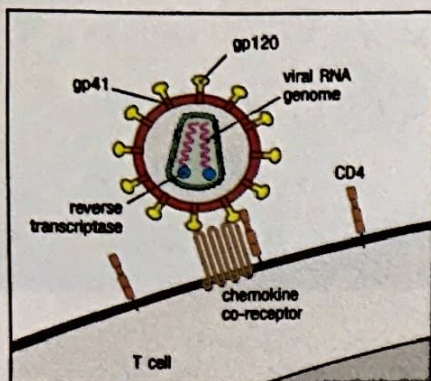


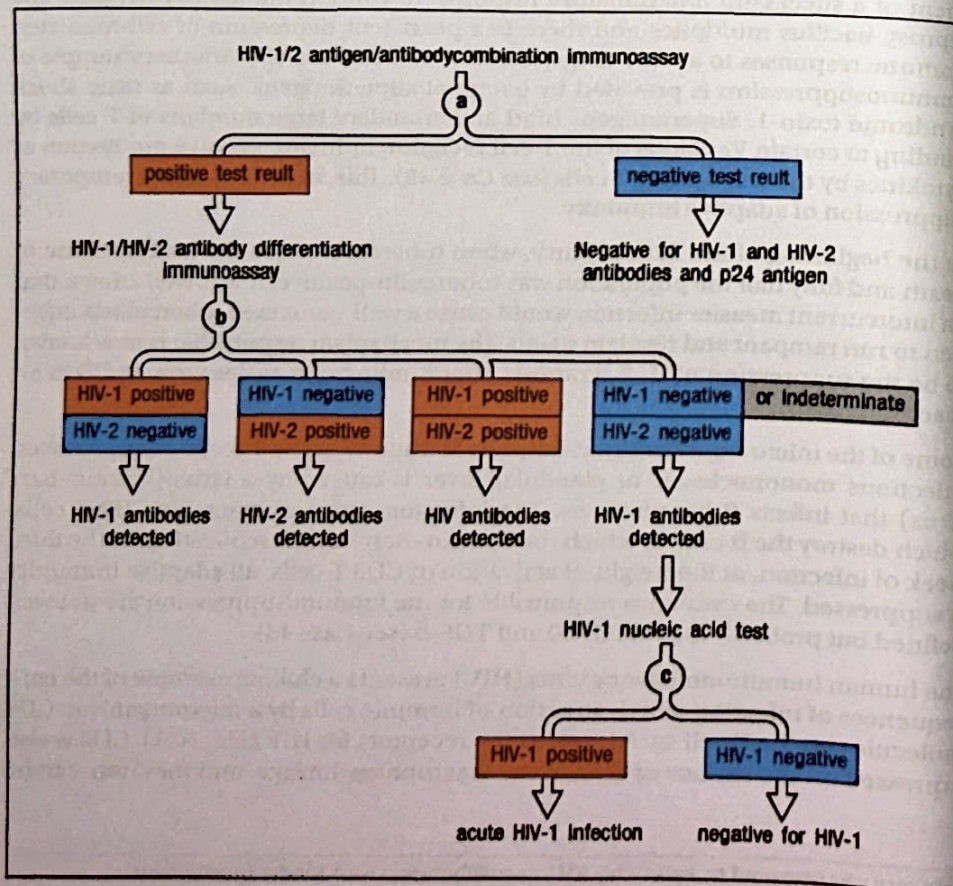
Fig. 10.1 HIV binds to CD4 T cells through its coat glycoprotein gp120. The gp120 molecule on the surface of the virus binds CD4 on T cells and macrophages; the viral protein gp41 then mediates fusion of the enveloped virus with the target cells, allowing the viral genome to enter the cell. The chemokine receptors CCR5 and CXCR4 act as co-receptors of HIV.

infected by this virus. The chemokine receptors CCR5 and CXCR4 act as obligatory co-receptors for HIV. As we shall see, the primary infection with HIV may go unnoticed, and the virus may replicate in the host for many years before symptoms of immunodeficiency are seen. During this period of clinical latency, the level of virus in the blood and the number of circulating CD4 cells remain fairly steady, but in fact both virus particles and CD4 cells are being rapidly destroyed and replenished as rounds of virus replication take place in newly infected cells. When the rate at which CD4 cells are being destroyed exceeds the capacity of the host to replenish them, their number decreases to a point at which cell-mediated immunity falters. As we have seen in other cases, such as severe combined immuno-deficiency (see Case 5), the failure of cell-mediated immunity renders the host susceptible to fatal opportunistic infections.

The case of Martin Thomas: a police officer whose past comes back to haunt him.

Martin Thomas is a 42-year-old police officer, who has always been in good health. He has been married for 10 years and has one child, an 8-year-old daughter. Six months ago he went to the emergency room at the local hospital complaining of a fever and a swollen right hand. He was admitted to the hospital for the hand infection, which was assumed to be the result of a cat scratch. His blood lymphocyte count was found to be very low, so a blood sample was sent to be tested for antibodies against the human immunodeficiency virus (HIV). The testing was done according to the algorithm illustrated in Fig. 10.2. ELISA (enzyme-linked immunosorbent assay) revealed the presence of anti-HIV antibodies and p24 antigen. Further testing

Fig. 10.2 Recommended laboratory HIV testing algorithm for serum or plasma specimens. a. Initial testing for HIV is performed with an FDA-approved antigen/antibody combination immunoassay that detects HIV-1 and HIV-2 antibodies and HIV-1 p24 antigen. This screens both for an established infection with HIV-1 or HIV-2 and for acute HIV-1 infection. No further testing is required for specimens that are nonreactive on the initial immunoassay. b. Specimens with a reactive result in (a) are tested with an FDA-approved antibody immunoassay that differentiates HIV-1 antibodies from HIV-2 antibodies. Reactive results on the initial antigen/antibody combination immunoassay (in a) and the HIV-1/HIV-2 antibody differentiation immunoassay (in b) are interpreted as positive for HIV-1 antibodies, HIV-2 antibodies, or HIV antibodies, undifferentiated. c. Specimens that are reactive on the initial immunoassay but nonreactive or indeterminate on the antibody differentiation immunoassay are tested with an FDA-approved HIV-1 nucleic acid test (NAT). A reactive HIV-1 NAT result indicates the presence of HIV-1 infection. A negative HIV-1 NAT result indicates a false-positive result on the initial immunoassay. (Courtesy of Centers for Disease Control and Prevention.)



using an FDA-approved immunoassay that differentiates HIV-1 from HIV-2 antibodies was positive for HIV-1 antibodies, indicating the presence of an HIV-1 infection. The results were reported to the patient's physician and the state public health authorities, according to the guidelines in Fig. 10.3. Officer Thomas was referred to Dr Wright, an AIDS specialist, at the Massachusetts General Hospital.

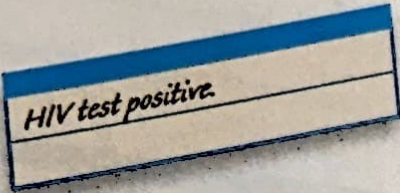
Mr Thomas told Dr Wright that he had had several sexual encounters with both women and men before his marriage 10 years ago. He had always been in good health until 6 months before the present consultation, when he began to have drenching night sweats several times a week. Over this period his body weight had gone down from 94.5 kg to 90 kg. He could not remember having any infections other than the one in his hand, nor any rashes, gastrointestinal problems, cough, shortness of breath, or any other symptoms. His mother had been 84 years old when she died of a heart attack, and his father had died at age 87 from cirrhosis of the

42-year-old man with a cat scratch that will not heal, and fever.

*Lymphocyte count very low.
Test for HIV.*

Test performed	Test results	Final interpretation for provider report	Test results to be reported to public health authorities
1. HIV-1/2 Ag/Ab combination immunoassay	1. Nonreactive	Negative for HIV-1 antigen and HIV-1/HIV-2 antibodies No laboratory evidence of HIV infection If acute HIV infection is suspected, consider testing for HIV-1 RNA.	Reporting this test result is not required
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay	1. Reactive 2. HIV-1 reactive and HIV-2 nonreactive	Positive for HIV-1 antibodies. Laboratory evidence consistent with established HIV-1 infection is present.	Report test results 1 and 2
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay	1. Reactive 2. HIV-1 nonreactive and HIV-2 reactive	Positive for HIV-2 antibodies. Laboratory evidence of HIV-2 infection is present.	Report test results 1 and 2
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay 3. HIV-1 nucleic acid test	1. Reactive 2. Nonreactive or indeterminate 3. HIV-1 RNA not detected	HIV antibodies were not confirmed and HIV-1 RNA was not detected. No laboratory evidence of HIV-1 infection. Follow-up testing for HIV-2 should be performed if clinically indicated.	Reporting this test result is not required
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay 3. HIV-1 nucleic acid test	1. Reactive 2. Nonreactive 3. HIV-1 RNA detected	Positive for HIV-1. Laboratory evidence consistent with acute HIV-1 infection is present.	Report test results 1, 2, and 3
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay 3. HIV-1 nucleic acid test	1. Reactive 2. Nonreactive 3. HIV-1 RNA detected	Positive for HIV-1 antibodies. Laboratory evidence of HIV-1 infection confirmed by HIV-1 RNA.	Report test results 1, 2, and 3
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay	1. Reactive 2. HIV-1 and HIV-2 reactive	Positive for HIV antibodies. Laboratory evidence of HIV infection is present. HIV antibodies could not be differentiated as HIV-1 or HIV-2. Additional testing for HIV-1 RNA or HIV-2 RNA should be performed if clinically indicated.	Report test results 1, and 2
1. HIV-1/2 Ag/Ab combination immunoassay 2. HIV-1/HIV-2 antibody differentiation immunoassay	1. Reactive 2. Nonreactive or indeterminate	HIV-1 antibodies were not confirmed and HIV-1 RNA testing was not performed. Testing of this specimen is incomplete. Follow-up testing for HIV antibodies and HIV-1 RNA is recommended as soon as possible.	Report test results 1, and 2

Fig. 10.3 Reporting results from the HIV testing algorithm to ordering health professionals and state authorities. The tests listed in the first column are the tests described in the flowchart in Fig. 10.2. Ag/Ab, antigen/antibody. (Adapted from Interim Guidelines for Laboratories on the Use of a New Diagnostic Testing Algorithm for Human Immunodeficiency Virus (HIV) Infection. Courtesy of New York State Department of Health.)



HIV test positive.

liver, cause unknown. His wife and child were both in good health and his wife, pregnant with their second child, had recently tested negative for anti-HIV antibodies. Mr Thomas told Dr Wright that he did not smoke or use intravenous drugs. He drank large amounts of beer on weekends. A cat and a dog were the only pets in the house.

On physical examination his blood pressure was 130/90, his pulse rate 92, and temperature 37.5°C (all normal). Nothing abnormal was found during the physical examination. His white blood cell count was $5800 \mu\text{l}^{-1}$ (normal), his hematocrit was 31.3 (slightly low), and his platelet count was $278,000 \mu\text{l}^{-1}$ (normal). His CD4 T-cell count was low at $170 \mu\text{l}^{-1}$ (normal 500–1500 μl^{-1}) and his HIV-1 RNA viral load was 67,000 copies μl^{-1} .

Mr Thomas was prescribed trimethoprim-sulfamethoxazole for prophylaxis against *Pneumocystis jirovecii* pneumonia (see Case 5). He was also given a combination antiretroviral agent consisting of tenofovir (TDF), emtricitabine (FTC), and efavirenz (Sustiva). He was counseled not to have sexual contact with his wife until his HIV viral load was suppressed, and to always use barrier protection thereafter. Because of her pregnancy, he was asked to immediately share his diagnosis with his wife and for her to see her obstetrician and an infectious disease specialist urgently to be HIV tested by viral load. She did so and was negative.

After 5 weeks of this therapy, Mr Thomas's HIV-1 viral load declined to 400 copies of RNA μl^{-1} and after 8 weeks to <20 copies of RNA μl^{-1} , in other words, to suppressed levels. In the meantime his CD4 T-cell count rose to 416 μl^{-1} . The prophylaxis for *Pneumocystis* was discontinued. Mr Thomas remains well and active and works full time; his wife delivered a healthy baby boy and remains uninfected.

Acquired immune deficiency syndrome (AIDS).

AIDS is caused by the human immunodeficiency virus (HIV), of which there are two known types, HIV-1 and HIV-2. HIV-2 was largely confined to West Africa but is now spreading into Southeast Asia and, with globalization, is found in many locales in low numbers. HIV infections in North and South America and in Europe are predominately due to HIV-1. HIV can be transmitted by vaginal, anal, or oral sexual intercourse, by infusion of contaminated blood or blood products, or by contaminated needles, which can be the major source of infection among drug addicts. The infection can also be passed from mother to child during pregnancy, during delivery, or by breastfeeding. In the past, between 25% and 35% of infants born to HIV-positive mothers were infected, but the rate of vertical transmission when women are offered suppressive antiretroviral therapy is <5% and if women have a suppressed HIV viral load, transmission is <2%.

Contact with the virus does not necessarily result in infection, but infection rates are associated with frequency of contact, route of contact (direct blood contact being the highest), and the amount of virus in the contacted bodily fluid. The standard indicator of infection is the presence of antibodies against the virus, and is now frequently detected by testing for HIV-1 and HIV-2 antibodies and HIV p24 antigen in a single assay. The initial infection, as in Mr Thomas's case, may pass unnoticed and without symptoms. More often, a mild viral illness within 6 weeks of infection is sustained, with fever, swollen lymph nodes, and a rash. It subsides at about the time that seroconversion (the appearance of anti-HIV antibodies) occurs, and although virus and antibody persist, the patient feels well. A period of clinical latency lasting years, and perhaps even decades, may ensue, during which time the infected person feels perfectly well. Then they may begin to experience low-grade fever, night sweats, excessive fatigue, fungal skin infections, or candidiasis (thrush) in the mouth or vagina. Lymph nodes in the neck, axillae (armpits), or groin may swell. Weight loss may become very marked. These are the prodromal

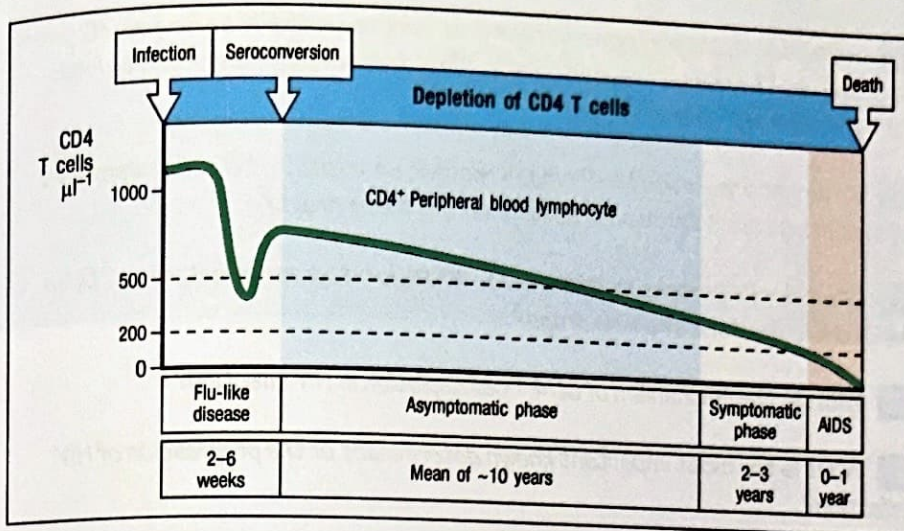


Fig. 10.4 The typical course of infection with HIV. The first few weeks are typified by an acute influenza-like viral illness, with high titers of virus in the blood. An adaptive immune response follows, which controls the acute illness and largely restores CD4 T-cell levels but does not eradicate the virus. Opportunistic infections and other symptoms become more frequent as the CD4 T-cell count falls, starting at around 350 cells μl^{-1} . The disease then enters the symptomatic phase.

symptoms of impending AIDS. (A prodrome is a concatenation of signs and symptoms that predict the onset of a syndrome.) The number of CD4 T cells in the blood may have been normal up to this time, but with the onset of the prodrome, the CD4 T-cell count begins to fall (Fig. 10.4). When the number of CD4 T cells decreases to the range of 200–400 cells μl^{-1} , the final phase of the illness, which is called AIDS, starts. At this time serious, eventually fatal, opportunistic infections as well as certain unusual malignancies occur (Fig. 10.5).

At any time after the infection, HIV may infect megakaryocytes, which have some surface CD4. Because megakaryocytes are the bone marrow progenitors of blood platelets, extensive infection of megakaryocytes causes the platelet count to fall (thrombocytopenia) and bleeding to occur. HIV may also infect the glial cells of the brain. Glial cells are of the monocyte-macrophage lineage and have some CD4 on their surface. The infection of glial cells may cause dementia and other neurological symptoms.

Many strides in the management and prevention of transmission of HIV infection have been made by early detection. It is strongly recommended that all pregnant women be tested with each pregnancy since antiretroviral therapy can suppress the HIV viral load and prevent transmission to the baby. It is also recommended that adolescents be tested at least once, as well as those in traditionally high-risk categories such as injection drug users and men who have sex with men.

Questions.

- 1 When Mr Thomas was first seen by Dr Wright, a chest X-ray revealed some enlarged hilar lymph nodes. If a lymph-node biopsy had been obtained, in what way would its histopathology have differed from that of lymph nodes from patients with severe combined immunodeficiency (SCID; see Case 5) or X-linked agammaglobulinemia (XLA; see Case 1)?
- 2 The course of an HIV infection in adults is very different from that in an infant infected in utero or intrapartum (during birth). What are the major differences between pediatric and adult AIDS, and how do you account for them?
- 3 What are the mechanisms of resistance to the progression of HIV infection?

Infections	
Parasites	<i>Toxoplasma</i> spp. <i>Cryptosporidium</i> spp. <i>Leishmania</i> spp. <i>Microsporidium</i> spp.
Bacteria	<i>Mycobacterium tuberculosis</i> <i>Mycobacterium avium</i> <i>intracellulare</i> <i>Salmonella</i> spp.
Fungi	<i>Pneumocystis jirovecii</i> <i>Cryptococcus neoformans</i> <i>Candida</i> spp. <i>Histoplasma capsulatum</i> <i>Coccidioides immitis</i>
Viruses	Herpes simplex Cytomegalovirus Herpes zoster
Malignancies	
Kaposi's sarcoma (invasive) Non-Hodgkin's lymphoma, including EBV-positive Burkitt's lymphoma Primary lymphoma of the brain	

Fig. 10.5 A variety of opportunistic pathogens and cancers can kill AIDS patients. Infections are the major cause of death in AIDS; of these, respiratory infection with *Pneumocystis jirovecii* is the most prominent. Most of these pathogens require effective macrophage activation by CD4 T cells or effective cytotoxic T cells for host defense. Opportunistic pathogens are present in the normal environment but cause severe disease primarily in immunocompromised hosts, such as AIDS patients and cancer patients. AIDS patients are also susceptible to several rare cancers, such as Kaposi sarcoma, non-Hodgkin's lymphoma, and EBV-associated Burkitt's lymphoma, suggesting that immune surveillance by T cells may normally prevent such tumors.

Case 10: Acquired Immune Deficiency Syndrome (AIDS)

- 4** A few individuals are known to have been infected with HIV as long as 20 years ago, and yet they remain asymptomatic. What factors may contribute to long-term survival with this infection?
- 5** Dr Wright prescribed Mr Thomas tenofovir, emtricitabine, and efavirenz as a combination tablet taken once daily. What are these drugs?
- 6** What do HIV protease inhibitors do? What do integrase inhibitors do? Does Mr Thomas need one of these drugs?
- 7** What is the mechanism of CD4 T-cell depletion in HIV infection?
- 8** What is the most important known determinant of the progression of HIV infection?
- 9** Which cytokine released during an HIV infection causes weight loss?

CASE 11

GRAFT-VERSUS-HOST DISEASE

Alien T cells react against their new host.

Bone marrow transplantation has proved to be useful therapy for some forms of leukemia, bone marrow failure (aplastic anemia), and primary immunodeficiency diseases. More recently, other sources of hematopoietic stem cells, such as peripheral blood stem cells and cord blood, have also been used for these purposes. Bone marrow and most other sources of hematopoietic stem cells also contain mature T lymphocytes, which may recognize the tissues of their new host as foreign and cause a severe inflammatory disease in the recipient. This is known as graft-versus-host disease (GVHD) and in its acute form it is characterized by a rash, which often starts on the face (Fig. 11.1), diarrhea, pneumonitis (inflammation in the lung), and liver damage. Fibrosis characterizes the chronic form of GvHD, which may affect various organs (skin, lungs, liver, gut, glands).

To achieve successful engraftment of bone marrow and avoid rejection of the transplant by the host, the immune system of the recipient must be destroyed and the recipient rendered immunoincompetent. This is usually accomplished with lethal doses of radiation or the injection of radiomimetic drugs such as busulfan, and the use of immunosuppressive drugs (cyclophosphamide, fludarabine). In children with severe combined immunodeficiency, who cannot produce T lymphocytes (see Case 5), this preparative treatment is not needed.

GVHD reflects reactivity of donor-derived T cells against the recipient's HLA antigens. GVHD occurs not only when there is a mismatch of classical MHC class I or class II molecules but also in the context of disparities in minor histocompatibility antigens; such minor differences are likely to be present in all donor-recipient pairs other than identical twins, even when donor and recipient are HLA-matched. Mature CD4⁺ T cells in the graft that are activated by allogeneic molecules produce a 'cytokine storm' that recruits other T cells, macrophages, and natural killer (NK) cells to create the inflammation characteristic of GVHD.

GVHD is arbitrarily called 'acute' if it occurs less than 100 days after hematopoietic stem cell transplantation, and 'chronic' if it develops after 100 days. Chronic GVHD differs from acute GVHD in other respects and is a more severe and difficult-to-treat problem. In the past, the presence of alloreactive T cells in the donor bone marrow was investigated by performing the mixed leukocyte reaction (MLR) (see Fig. 6.3), in which lymphocytes from the potential donor are mixed with irradiated peripheral blood mononuclear cells from the potential recipient. If the donor

TOPICS BEARING ON THIS CASE:

T-cell recognition of nonself MHC molecules

Reactions of T cells against foreign antigens

Transplantation

Minor histocompatibility antigens

Mixed lymphocyte reaction